Ekarsi Lodh

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Research Profile

Computational researcher with a BTech in Computer Science and Engineering and an MSc in Bioinformatics from the University of Birmingham (2026); currently a Postgraduate Teaching Assistant and a Research Associate in the Department of Cancer and Genomic Sciences, benchmarking fine-tuned large language models (InternistAI-7B, Gemma-2-9B) for clinical prediction on EHR data with QLoRA/PEFT. First author of six peer-reviewed papers, including in the *Journal of Computer-Aided Molecular Design*, *Journal of Molecular Graphics and Modelling*, and *Computational Intelligence*, spanning deep learning for protein–ligand binding, GPU-accelerated sequence alignment, and reinforcement learning for drug prioritisation. Experienced in building reproducible Python and R pipelines for genomics, epigenomics and healthcare data on HPC (BlueBEAR), and in teaching and mentoring computational research.

Education

2025 – 2026 MSc in Bioinformatics, **University of Birmingham, UK** (**Distinction**; Graduation: Dec 2026)
Thesis: *Epigenetic Age Acceleration in Early-Onset Breast Cancer: Multi-Cohort and Cross-Omic Analysis* (Supervisor: Dr Archana Sharma-Oates)

2020 – 2025 BTech in Computer Science and Engineering, **Techno Main Salt Lake, India** (CGPA: 8.13/10.0)
Thesis: *A Stacked Ensemble Meta-Classifier Framework for PAM50 Breast Cancer Subtype Prediction from Gene Expression Signatures* (Supervisor: Dr Tapan Chowdhury; published as [J6])

Research Experience

Research Associate Mar 2026 – Present
Department of Cancer and Genomic Sciences, College of Medicine and Health, University of Birmingham, UK (Supervisor: Dr Mamunur Rashid)

- Benchmarking zero-shot and fine-tuned LLMs (**InternistAI-7B**, **Gemma-2-9B**) for clinical decision support on three longitudinal EHR datasets – MIMIC-IV (364,627 patients), Synthea (∼575,000 synthetic patients) and a Delphi-2M synthetic UK Biobank cohort – for mortality, 14-day readmission and next-diagnosis prediction, with ethnicity- and sex-stratified fairness evaluation.
- Fine-tuning with **QLoRA** (4-bit NF4; full- vs last-layer adapters) on GPU HPC and comparing full-history vs recent-visit prompting against XGBoost, DoctorAI, Delphi-2M and Longformer baselines.
- Fine-tuning raised InternistAI's MIMIC-IV mortality **AUROC from 0.678 (zero-shot) to 0.946**, above XGBoost (0.939); fine-tuned LLMs achieved the highest AUROC on all three readmission datasets (MIMIC-IV: 0.766 vs 0.736 for XGBoost).

MSc Thesis Project Nov 2025 – Aug 2026
School of Biosciences, University of Birmingham, UK (Supervisor: Dr Archana Sharma-Oates)

- Investigated epigenetic age acceleration and earlier breast cancer onset in UK-resident South Asian women using blood-derived DNA methylation data across multiple cohorts.
- Integrated five epigenetic clocks (Horvath, Hannum, PhenoAge, GrimAge, DunedinPACE) with robust preprocessing, cell-composition adjustment and multi-clock triangulation to assess associations with breast cancer risk and age at diagnosis.
- Built reproducible statistical and ML workflows in R and Python, with a focus on under-represented populations.

Group Research Project – Polyploid GWAS Oct 2025 – May 2026
School of Biosciences, University of Birmingham, UK (Supervisor: Dr Lindsey Compton)

- Optimised GWAS pipelines in R and Linux to identify loci underlying canopy architecture and yield in autotetraploid potato from SNP and phenotyping data.
- Integrated population structure analysis (DAPC, STRUCTURE), environmental correction with BLUEs, and comparison of statistical models to improve trait–locus associations.

Research Intern Feb 2023 – Sep 2025
Dept. of Computer Science and Engineering, Techno Main Salt Lake, India (Supervisor: Dr Tapan Chowdhury)

- Developed deep learning models for protein–ligand binding affinity prediction (RLBindDeep, CGDeepAff), a GPU-accelerated global sequence alignment framework (BioAlignNet) and network-informed offline RL for pharmacogenomic drug prioritisation (NetPolicy-RL).
- Carried out signalling pathway modelling, molecular docking and drug repurposing studies; this work underpins most of the publications listed below.

Publications

Own name in bold. First-author publications marked with *.

Peer-Reviewed Journal Articles

[J1] A. Ray, K. Agarwal, S. Jha, A. M. Singh, S. Majumder, **E. Lodh**, and T. Chowdhury, "miRNA–mRNA interaction

network analysis in Alzheimer's disease for biomarker discovery," *Journal of Molecular Neuroscience*, vol. 76, no. 123, Aug. 2026. doi: 10.1007/s12031-026-02579-w

[J2] T. Chowdhury, **E. Lodh**, and S. Majumder, "A Novel Method for Drug Application on Signaling Pathways with Concurrent Faults Using Probabilistic Boolean Networks," *Computational Intelligence*, vol. 42, no. 4, e70276, Jul. 2026. doi: 10.1111/coin.70276

[J3] * **E. Lodh**, S. Majumder, T. Chowdhury, and M. De, "NetPolicy-RL: Network-Informed Offline Reinforcement Learning for Pharmacogenomic Drug Prioritization," *Journal of Computer-Aided Molecular Design*, vol. 40, no. 140, Jun. 2026. doi: 10.1007/s10822-026-00823-4

[J4] S. Majumder, **E. Lodh**, T. De, and T. Chowdhury, "In silico Characterization of Melanoma-derived Mouse lncRNA Gm26982 and Comparative Analyses with Human Ortholog," *Molecular Genetics and Genomics*, vol. 301, no. 103, Apr. 2026. doi: 10.1007/s00438-026-02423-1

[J5] * **E. Lodh**, T. Chowdhury, and S. Majumder, "BioAlignNet: A GPU-Accelerated Framework for Efficient Global Sequence Alignment Across Genomics and Proteomics Scales," *Computational Intelligence*, vol. 42, no. 3, e70232, Apr. 2026. doi: 10.1111/coin.70232

[J6] * **E. Lodh**, T. Chowdhury, A. Shaw, D. Bedajna, S. Bera, S. Majumder, and M. De, "A Stacked Ensemble Meta-Classifier Framework for PAM50 Breast Cancer Subtype Prediction from Gene Expression Signatures," *Network Modeling Analysis in Health Informatics and Bioinformatics*, vol. 15, no. 78, Mar. 2026. doi: 10.1007/s13721-026-00755-x

[J7] * **E. Lodh**, S. Majumder, T. Chowdhury, and M. De, "RLBindDeep: A ResNet-LSTM Based Novel Framework for Protein–Ligand Binding Affinity Prediction," *Journal of Molecular Graphics and Modelling*, vol. 144, 109282, Jan. 2026. doi: 10.1016/j.jmgm.2026.109282

[J8] T. Chowdhury, S. Basu, A. Roy, V. Sarkar, K. Mondal, **E. Lodh**, and M. De, "Graph-theoretic and emotional analysis of character dynamics in novel-to-film adaptations," *Journal of Computational Social Science*, vol. 9, no. 18, Jan. 2026. doi: 10.1007/s42001-025-00451-2

[J9] S. Majumder, **E. Lodh**, and T. Chowdhury, "Implications of trinodal inhibitions and drug repurposing in MAPK pathway: A putative remedy for breast cancer," *Computational Biology and Chemistry*, vol. 113, 108255, Oct. 2024. doi: 10.1016/j.compbiolchem.2024.108255

Peer-Reviewed Conference Papers (P = presenter)

[C1] * **E. Lodh**, S. Majumder, T. De, T. Chowdhury, and M. De, "GeneDeepNet: A Differential Expression and Deep Learning Based Novel Framework for Detecting Invasive Breast Cancer Subtypes," *ICMLDE-2025, Procedia Computer Science*, vol. 283, pp. 1235–1248, 2026. doi: 10.1016/j.procs.2026.06.202 (P)

[C2] * **E. Lodh**, S. Majumder, and T. Chowdhury, "CGDeepAff: Deep Learning-Based Approach for Protein–Ligand Binding Affinity Estimation Using CNN-GRU," *8th IEMENTech*, Kolkata, India, 2025, pp. 1–6. doi: 10.1109/IEMENTech65115.2025.10959578 (P)

[C3] T. Chowdhury, U. Das, A. Makur, A. Sinha, D. Jana, A. Das, and **E. Lodh**, "Ensemble Clustering on Big Data," *8th IEMENTech*, Kolkata, India, 2025, pp. 1–6. doi: 10.1109/IEMENTech65115.2025.10959554 (P)

[C4] M. De, R. L. Chhetri, M. Konar, A. N. Joardar, A. Jain, **E. Lodh**, and T. Chowdhury, "A Deep Learning-Based Method for the Categorization of Different Skin Diseases," *AdComSys 2025, Lecture Notes in Networks and Systems*, vol. 1887, pp. 40–56, Springer, Cham. doi: 10.1007/978-3-032-20253-6_4 (P)

Technical Skills

Programming	Python, R, Bash/Shell scripting, Awk, C; Linux command line
Machine Learning & AI	PyTorch, TensorFlow, Hugging Face Transformers; LLM fine-tuning (QLoRA, PEFT, 4-bit quantisation); BioClinicalBERT and domain-specific transformer adaptation; LSTM/GRU/CNN/ResNet architectures; ensemble methods, XGBoost; offline reinforcement learning; evaluation (AUC-ROC, precision, recall, F1)
Bioinformatics & Omics	DNA methylation and epigenetic clocks; GWAS and population structure (DAPC, STRUCTURE, BLUEs); NGS and structural-variant (VCF) analysis; gene expression and differential expression; sequence alignment; network and pathway analysis (Cytoscape); molecular docking (AutoDock Vina, PyRx, UCSF Chimera, ClusPro); ncRNA characterisation (RNAfold, CPC2, LncBase)
HPC & Software	BlueBEAR HPC (large-scale genomics workflows; GPU-accelerated deep learning and LLM fine-tuning); Streamlit web apps; R Markdown; reproducible command-line pipelines

Selected Software & Projects

Clinical Mortality Prediction with Fine-Tuned Clinical LLMs 2026
MSc Machine Learning module

– Built in-hospital mortality prediction models on MIMIC-style structured EHR data; fine-tuned Internist-7B with QLoRA (4-bit PEFT) and fully fine-tuned BioClinicalBERT, benchmarked against XGBoost, LSTM and transformer baselines (AUC-ROC, precision, recall, F1).

Delly Trio SV Analyzer – CLI pipeline and Streamlit web app 2025

MSc Genomics & NGS module (pipeline, app code, live app)

- End-to-end pipeline for QC, de novo calling, annotation, pathogenicity tiering and plotting of structural variants from Delly trio VCFs, producing structured tables and reports for downstream use.
- Deployed as an interactive Streamlit app with trio role inference, X-chromosome heterozygosity-based sex inference and Mendelian-consistency checks for de novo variants.

Reproducible Statistical Analysis Pipeline in R 2026
MSc Mathematics & Statistics module (GitHub)

- Modular R and R Markdown pipeline for hypothesis testing, distributional analysis and mathematical modelling; graded 78.5%.

Teaching & Mentoring

Postgraduate Teaching Assistant – MSc Bioinformatics Sep 2026 – Present
University of Birmingham, UK

- Supporting practical sessions in Linux and command-line computing, Python and R, NGS pipelines, statistical machine learning and health data analytics; troubleshooting students' code and analytical workflows.

Undergraduate Research Project Co-Supervisor Aug 2025 – Present
Techno Main Salt Lake, India (with Dr Tapan Chowdhury)

- Remotely co-supervising BTech final-year students on research design and computational methods; guided a miRNA–mRNA network analysis for Alzheimer's disease biomarker discovery that resulted in a student-led article in *Journal of Molecular Neuroscience* [J1].

Leadership, Service & Memberships

Mar 2026	AI Hackathon Volunteer, **HealthTech AI Hub, University of Birmingham** – supported an international healthcare AI hackathon; evaluated an NLP model translating Bengali medical check-up data into English for accuracy and clinical relevance
2025 – 2026	Postgraduate Student Ambassador and MSc Bioinformatics Student Representative, **University of Birmingham**
2025 – Present	Member, **IEEE** (Computational Intelligence Society; Engineering in Medicine and Biology Society)
2023 – 2025	Departmental Convenor (2024–25) and Class Representative, Dept. of CSE, **Techno Main Salt Lake**
2021	1st place, LOKNITI Policy Conclave, Public Policy & Opinion Cell, **IIT Kanpur**

Training, Certifications & Languages

Ethics & Governance	CITI Program – Data or Specimens Only Research (human subjects research training required for PhysioNet credentialed access to MIMIC-IV); research data management and information governance training, University of Birmingham
Professional Training	Postgraduate Teaching Assistant induction, University of Birmingham; BlueBEAR High-Performance Computing training, University of Birmingham
Languages	English: IELTS 8.0 overall, CEFR C1 (2025); German: Goethe-Zertifikat A1, 71/100 (2024)
Certifications (Coursera)	Bioinformatics for Beginners, UC San Diego (2025); Python for Genomic Data Science, Johns Hopkins (2024); Introduction to Machine Learning, Duke (2023); Introduction to Python Programming, UPenn (2022)